

Received 11 February 2025, accepted 4 March 2025, date of publication 10 March 2025, date of current version 8 April 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3549440

## RESEARCH ARTICLE

# A Hybrid Brain Tumor Classification Using FL With FedAvg and FedProx for Privacy and Robustness Across Heterogeneous Data Sources

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This work was supported by the Deanship of Postgraduate Studies and Scientific Research at Majmaah University under Project R-2025-1641.

**ABSTRACT** Data privacy and heterogeneity among healthcare settings present fundamental challenges to machine learning (ML) brain tumor classification (BTC) model development based on local data. In this paper, we outline the need to develop an effective brain tumor diagnosis model while ensuring data security. We propose a distributed learning framework based on Federated Learning (FL) to build such a system. This framework combines Federated Averaging (FedAvg) and Federated Proximal (FedProx) to train Convolutional Neural Networks (CNN) on data hosted by multiple clients. FedAvg saves these locally trained models from the patients to make a global model while keeping sensitive patient data decentralized and private. Nevertheless, traditional FL suffers from model instability and divergence due to the disparity of data distributions among clients. FedProx addresses this challenge by introducing a proximal regularization term that penalizes divergence from the global model to stabilize local training considering data heterogeneity. In this work, we utilized the Brain Tumor MRI Dataset from Kaggle and trained the model to classify four brain tumors (BT), namely glioma, meningioma, pituitary tumor, and no-tumor class. Experiments were conducted for ten communication rounds, with each client training the model independently based on its local dataset and without sharing data. Results revealed that the method consistently outperformed accuracy, recall and F1-score, further validating the capabilities of the proposed framework in effectively handling data heterogeneity while conserving data privacy. The global model outperformed models that had been centralized and trained on a smaller, common dataset observed in FL scenarios to improve the potential for collaborative medical diagnosis. This method is especially suitable for medical applications with strict privacy regulations, as it provides an approach to linking heterogeneous data from multiple healthcare institutions.

**INDEX TERMS** Federated learning, brain tumor classification, FedAvg, FedProx, data heterogeneity, privacy protection, machine learning in healthcare, medical imaging.

The associate editor coordinating the review of this manuscript and approving it for publication was Zhen Ren<sup>1</sup>.

## I. INTRODUCTION

Brain tumors (BT) are a high-mortality public health problem that consists of heterogeneous tissues originating from

different anatomical regions of the brain. Primary BT affects about 14 persons per 100,000 people annually. More common is the second BT, which puts more pressure on healthcare systems worldwide [1]. BT not only influences physical health but also plays an important role in the quality of life, cognitive functions, and emotions of the patients [4], [5]. The heterogeneous nature and response to the treatment of brain tumors, among other clinically relevant factors, underlines the need for advancing diagnostic and therapeutic approaches to improve patient outcomes [3]. Recent major advances in medical imaging and AI have polished the way BT is identified and classified. Deep learning methods, specifically CNN, have been demonstrated to perform particularly well in estimating medical images. This has resulted in automated approaches for locating BT [7]. Using large databases of annotated medical images, these models learn the nuanced associations that are associated with different classes of tumors [5], which help improve diagnostic precision and efficiency. The nature of patient information is a major roadblock for healthcare organizations to cooperate and share data. However, FL (FL) can be the solution to such problems.

Federated Learning, a distributed Machine Learning approach, permits many healthcare agencies to work together to train model while enjoying the privacy of their patient's private data. This approach provides increased data privacy without losing the ability to develop strong prediction models. For BT segmentation and classification, FL enables data from multiple medical centers to be merged without compromising individual data privacy, thus enhancing the robustness and accuracy of the model [6], [9]. Jeffrey et al. [1] indicated that it would be acceptable, for example, to obtain high similarity between FL and centralized learning methods while preserving patient data privacy (i.e., decentralized infrastructure), hence making FL a promising approach in medical applications. In addition, FL models are designed specifically to address the unique challenges of BT analysis, such as data heterogeneity and segmentation task requirements for high accuracy [8].

Here, Figure 1 shows a high-level architectural overview of FL, specifically the communication flow between Clients and Central Server in terms of rounds during model training. Here is an outline of how the pieces come together:

1. **Clients:** Clients are individual devices or entities (like mobile phones, sensors, and computers) that all have their own local data. FL is a distributed data training approach that maintains the privacy of local data, where each client trains the model on its private data without directly providing it to a server.

2. **Server:** It is the centralized entity to manage the global model. Responsible for coordinating the training process by aggregating models after clients run their local training, updating the global model, and returning this updated model to clients.

3. **Global Model:** The global model held by the server. The aggregation of the locally trained models received from the clients updates this model.

4. **Aggregation:** It is the way the server aggregates updates sent from all clients. In the aggregation step, a weighted average or some other aggregation is calculated to obtain the updated global model using client contributions.

#### A. STEP BY STEP PROCESS

1. **Send model for training on the local data of the client:** In this first step, a server would send an initial version of the global model to each client. This model is then sent to each client device (an individual phone, sensor, or computer), which locally prepares for training on the new model.
2. **Local training of the model and sending it to the server:** Here, every client trains the model on its local data. In contrast to this local training, instead of raw data, the clients return model updates (often the learned weights or gradients) to the server. This means that no private data will ever leave client devices, thus protecting privacy.
3. **Aggregate the models received from all clients and then update the global model:** The server gathers all of the client model updates. It aggregates these updates.
4. **Send the updated model to clients and repeat:** At the same time, once the server alters the global model, the server transmits the updated model to the clients. This process goes on locally, then aggregates and is updated globally. It goes through many cycles known as rounds of communication until the model's performance is optimal.

This cyclic workflow permits the model to leverage multiple data sources without violating the data privacy of a particular dataset. The areas of the biggest benefit of FL are healthcare, finance, and individualized applications, as data security and privacy are highly important there. Because in FL, data are kept local on the device, and only the model updates are sent, the communication costs are also lower, besides getting more accurate models on clients with different data distribution.

#### B. RESEARCH OBJECTIVE

This research aims to:

1. Enhance the data confidentiality and privacy using FL to train the models with the multiple providers' data without insisting on centralizing patient data.
2. This makes use of the Federated Averaging (FedAvg) and Federated Proximal (FedProx) algorithms to address data heterogeneity issues for clients in the development of a globally efficient brain tumor classification model.
3. Moreover, it reviews the effectiveness of FedProx's proximal regularization in enhancing stability and accuracy in the case of non-IID data distribution.
4. Classify BT using CNN on decentralized data.

#### C. RESEARCH CONTRIBUTIONS

##### 1) INTRODUCTION OF A FL FRAMEWORK FOR BTC

Herein, this research proposes a distributed learning framework involving FedAvg and FedProx for CNN training

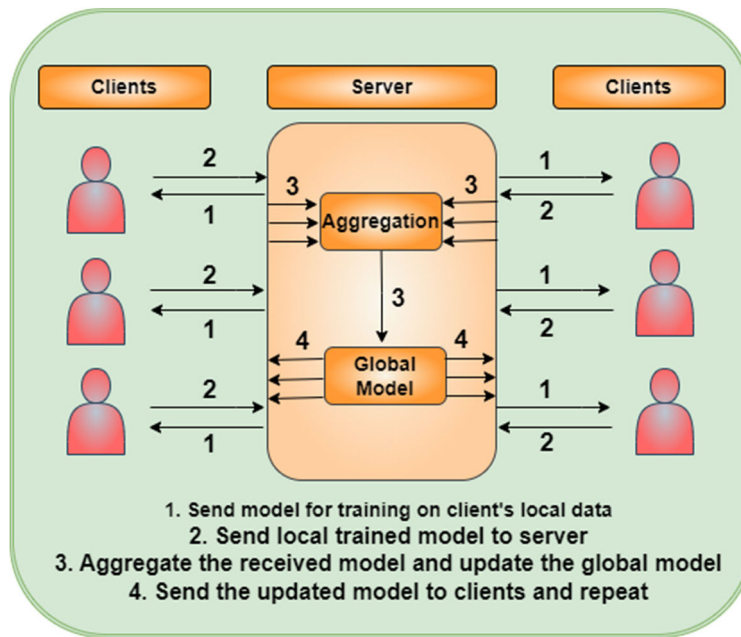


FIGURE 1. Federated learning process flow.

on data distributed across multiple healthcare providers. Another requirement for medical applications is avoiding direct sharing of data, and through FL, the framework achieves data privacy to allow collaborative model training.

## 2) INTEGRATION OF FEDPROX FOR HANDLING DATA HETEROGENEITY

Due to the fact that the distributions of data across multiple healthcare providers are inherently heterogeneous, the study uses FedProx to handle this heterogeneity. FedProx proposes to add a proximal regularization term that discourages the local model from deviating far from the global model during its training, which contributes to more robust convergence across non-IID data scenarios.

## 3) COMPREHENSIVE EVALUATION OF MODEL PERFORMANCE

The performance of the framework is then tested by carrying out experiments with ten communication rounds. The result shows that the combined approach of FedAvg and FedProx leads to consistent improvement in BTC.

## 4) PRESERVING DATA PRIVACY IN MEDICAL DIAGNOSIS

This work shows that the FL technique could promote the application of maintaining data privacy for medical diagnosis, with a capacity to seek an efficient and safe way of constructing a model cooperatively among numerous healthcare facilities. This is especially important in cases where privacy is of utmost importance, as it may be applied in medical imaging and healthcare analytics.

## 5) SUPERIOR PERFORMANCE OF GLOBAL MODEL

As illustrated in Fig. 6, the global model trained through the proposed framework outperforms the other models with a centralized learning approach, underscoring the effectiveness of FL in harnessing disparate and decentralized datasets. Our work demonstrates that in the case of non-IID data, the given framework allows patients to be classified into four classes of brain tumors. Therefore, our study has strong pragmatic significance.

## II. LITERATURE REVIEW

The classification of BT is a pivotal question in medical imaging, especially because of the tumor heterogeneity, which in turn can affect the therapeutic management and the prognosis of patients. The new DL algorithms, especially CNNs, can help perform BTC from MRI data automatically. For instance, Xie et al. [19] present a detailed study of CNN methodologies for BTC problems, and the issues and potential development in the field. They stress the importance of applying the principles related to such aspects as trust and transparency in the clinical uses of such models, as only a few works have considered the fact that CNNs are black boxes. To this family of works, Mohan et al. [18] present a new CNN-based medical image analysis and detection network compared with VGG16 net. These reveal that deep learning models are capable of differentiating between primary brain tumors and metastases with high sensitivity and specificity. Likewise, Johnpeter and Ponnuchamy [15] discussed a computer-aided detection of glioma tumors through CNN with optimistic classification accuracies, further enhancing the applicability of DL in this regard. Also, an amalgam of

architectures that employ different machine-learning approaches has been proposed. For instance, a combined feature extraction technique for BTC is described in [14] using MX-ML and regularised extreme learning, which points to the possibility of optimizing the performance of classification by combining different approaches. Additionally, Azzahra [12] employs Inception ResNet V2 for MRI-based BTC, with a view to applying the multi-class classification to get different types of tumors such as meningioma, glioma, and pituitary.

In the case of healthcare data privacy and sharing, it has been seen that FL is a possible paradigm. Due to this decentralized system, various hospitals can come together and train our models without any patient's personal health information being revealed. In the case of their study assessing the PySyft FL system, Budrionis et al. [13] recommend its use mainly in healthcare scenarios where data centralization is not always possible. Atha [11] puts forward SSBTCNet, which uses FL to improve model accuracy and protect patients' data. The combination of FL with DL techniques has proved to enhance the BTC accuracy in image analysis. For example, Liu [17] described the use of triplet contrastive learning for BTC; however, the method can be coupled with FL approaches to improve the model performance across datasets. Moreover, the results of this study reveal that more complex architectures, such as RNNs described by Kalaiselvi et al. [16], should be the target of the FL approach to improving the classification of the BT.

Furthermore, the FedProx algorithm is designed to tackle some problems of FL, especially when working with different kinds of data. This algorithm enables the incorporation of local updates at the same time as the global model; this is important given that institutional data distributions may differ greatly in the medical field. Aamir et al.'s [10] work discusses how FedProx improves the performance of FL models for BTC while adapting to different datasets with regard to data privacy. The study done by Hossain et al. [29] pays attention to the transition from binomial to multiclass classification and improves the prognosis by capturing the real-world model of distinguishing one type of disease from another. Their studies show that several deep learning frameworks like VGG16 with 95.11%, ResNet50 at 93.88%, and the proposed IVX16 model yielded the highest accuracy of 96.94%. Furthermore, the adoption of transfer learning in the IVX16 model is an example of the following state-of-the-art trend, in which using pre-trained networks helps to achieve high results even when the amount of data collected in healthcare is limited. Notably, the integration of Explainable AI into their workflow not only confirms the predictions formulated by the model but also solves the most significant problem of transparency in AI-based healthcare solutions. Such advancements show that DL could significantly transform BTC and detection, keep patients from adverse outcomes, and respect ethical data use [29].

To what extent these two methods can improve the precision of BT image processing was the focus of the study

conducted by Reddy and his group in 2024. The proposed approaches tested the research using MRI images of the tumor origin acquired from different types of BT; BWT and ROBPCA samples consisted of 16. Having completed image segmentation, the researchers derived textural features required for tumor work. Specifically, the test was mostly centered around just how accurate and sensitive or insensitive it was. It used a 1:1 enrolment ratio, a significance level of 5%, a desired power of 80%, and a confidence level of 95%. It was found that there was no difference between the two groups that were significant at the level of 0.05 ( $F = 2.493$ ;  $p = 0.182$ ). Combined, the sensitivity was 81.5%, and specificity was 84%, with the ROBPCA method being superior to BWT in the detection and analysis of tumor regions [28]. A CNN model was developed by Das et al. [30], which has the ability to diagnose brain tumors in T1-weighted contrast-enhanced MRI images. These results show why it is crucial to enhance the picture quality before the classification process is done. The authors of the paper worked with a dataset of the number of 3,064 pictures. The experiments of the CNN model were highly successful, with an average accuracy of 94.39 %, average precision of 93.33 %, and average recall of 93%. A similar performance suggests that the proposed model improves the diagnosis of brain tumors over current approaches due to the ability of CNNs to enhance classification tasks in the context of medical images [30].

### III. PROBLEM STATEMENT

Medical imaging, like MRI studies, is used to classify brain tumors. This is an important part of diagnosing and planning treatments for patients. Traditional machine learning models require centralized datasets, which can be challenging to achieve due to data privacy concerns, particularly in the medical field where patient data is highly sensitive. Hospitals and medical institutions often have isolated, decentralized datasets that cannot be easily shared, limiting the potential for collaborative learning from a larger, more diverse set of data. The goal is to use FL to make a strong BTC system that protects privacy. The main goal of this work is to use the FedAvg and FedProx methods to let many institutions train a DL model together without sharing their data. The research looks into how to quickly train a model that can sort four different kinds of brain conditions using data from different clients, taking into account the fact that these clients have different amounts of data. This work aims to address the following key challenges:

1. **Data Privacy:** Developing a system that allows collaboration without sharing sensitive medical data across institutions.
2. **Client Heterogeneity:** Handling differences in data distribution, size, and computational power across clients by incorporating the FedProx proximal term into the training process.
3. **Model Performance:** Achieving high accuracy, precision, recall, and F1-score in classifying brain tumors while training models in a decentralized environment.

**TABLE 1.** Various brain tumor model comparison.

| Ref  | Authors                         | Model                          | Advantages                                                                                                 | Disadvantages                                                                                      | Accuracy | Year |
|------|---------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------|------|
| [26] | Ullah et al.                    | Gabor Filter + ResNet50 + SVM  | Superior feature extraction, high accuracy, comprehensive dataset                                          | Complexity of hybrid model, high computational requirements                                        | 95.73%   | 2023 |
| [27] | M. Ravinder et al.              | CNN-based Graph Neural Network | Combines GNN and CNN for improved detection, accounts for non-Euclidean distances, high accuracy of 95.01% | Potential complexity in model training and tuning may require extensive computational resources    | 95.01%   | 2023 |
| [28] | Chappidi Sree Teja Reddy et al. | BWT                            | Simple to implement, effective in various imaging contexts                                                 | The BWT algorithm is highly sensitive to noise, which affects its performance                      | 81.50%   | 2024 |
| [28] | Chappidi Sree Teja Reddy et al. | ROBPCA                         | Better than the BWT algorithm in terms of accuracy                                                         | The ROBPCA algorithm outperforms BWT in accuracy, reaching 84%, but further improvement is needed. | 84%      | 2024 |
| [29] | S. Hossain et al.               | IVX16                          | High accuracy (96.94%) with the proposed model (IVX16). - Comparative analysis of several top DL models.   | The dataset size is relatively small for deep learning models (3264 images).                       | 96.94%   | 2024 |
| [30] | S. Das et al.                   | CNN                            | Achieved high accuracy (94.39%) and satisfactory performance.                                              | It may require further generalization for other types of tumors or larger datasets.                | 94.39%   | 2019 |

#### IV. PROPOSED METHODOLOGY

Figure 2 shows the proposed methodology and the process flow.

##### A. DATASET DESCRIPTION

This BTC task uses the Brain Tumor MRI Dataset from Kaggle, which is a combination of figshare, SARTAJ, and Br35H dataset images with 4 classifications such as gliomas, meningiomas, pituitary tumors, and no tumors. For model training and evaluation, the dataset is separated into training and testing subsets. In the training portion, 1,321 gliomas, 1,339 meningiomas, 1,457 pituitary tumors, and 1,595 no tumor photos make up 5,712 images. The testing subset includes 1,311 images: 300 gliomas, 306 meningiomas, 300 pituitary tumors, and 405 no tumors [31]. The dataset provides a balanced and comprehensive representation of brain tumor types and normal brain scans, ensuring the model has sufficient data to learn the distinguishing features between each class. The inclusion of both train and test data enables the evaluation of the model's generalization capability. The variety in the dataset allows for detailed performance analysis, particularly in FL scenarios, where data from multiple clients or sources can be aggregated while maintaining data privacy. This dataset is ideal for building a robust multi-class classification model capable of accurately diagnosing different types of BT from MRI images.

##### B. LOAD AND PRE-PROCESS DATA

To begin, load the MRI dataset of brain tumors. To make sure that the input dimensions are consistent, every image is shrunk to a standard size of  $128 \times 128$  pixels. Pixel values are normalized by scaling them to the range  $[0,1]$  in order to make model training more efficient. Additionally, class labels are converted to one-hot encoded vectors, enabling the model to perform multi-class classification. For instance, the class label for glioma is represented as  $[1,0,0,0]$  for a four-class problem.

##### C. SPLIT DATA AMONG CLIENTS

For the purpose of simulating an FL scenario, the dataset is divided into  $K$  parts, where  $K$  is the number of clients that are taking part in the training process. This split should ensure that each client receives approximately the same number of data, helping to maintain balance in the training distribution. This process mimics real-world scenarios where data is often distributed across multiple clients, such as hospitals or institutions.

##### D. CNN MODEL

A CNN is defined to classify the BT images. The architecture typically includes several convolutional layers followed by pooling layers and ReLU activation functions to introduce non-linearity. The final layer is a softmax layer that outputs

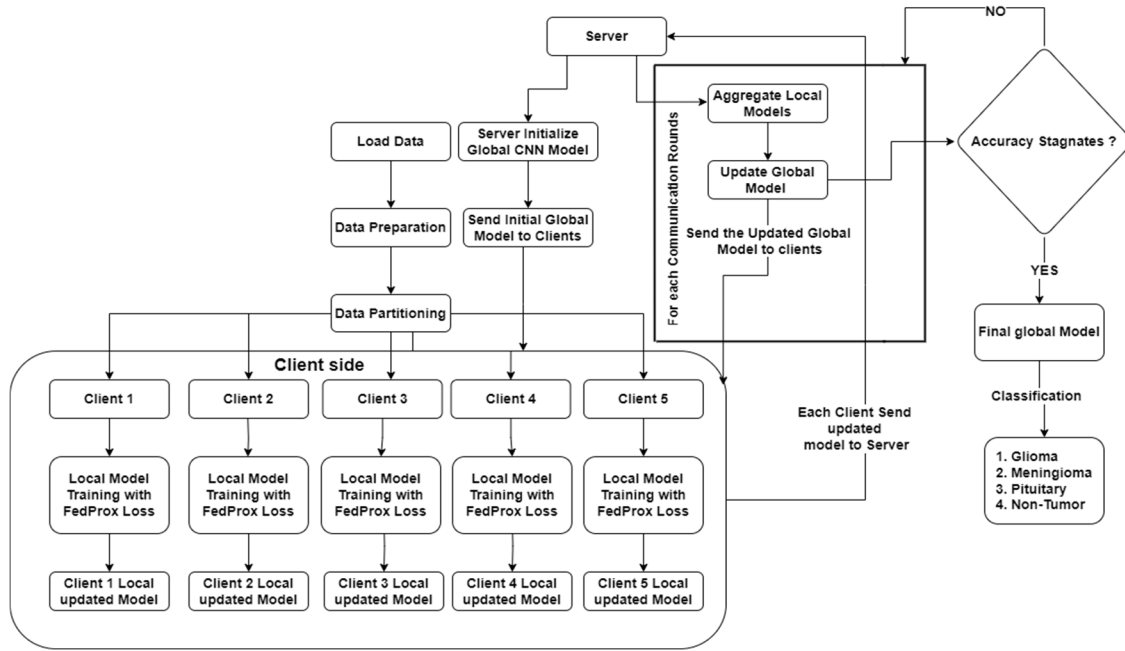


FIGURE 2. Flow of proposed methodology.

probabilities for the four categories, enabling the model to make predictions for the given input images

#### E. DEFINE FEDPROX LOSS

The FedProx loss function is designed to address the heterogeneity among clients in FL. It comprises two components: The local loss, which is computed using categorical cross-entropy:

$$\mathcal{L}_{local}(D_i, \omega_i) = - \sum_{j=1}^C y_{ij} \log(\hat{y}_{ij})$$

where  $C$  is the number of classes,  $y_{ij}$  is the true label for instance  $j$ , and  $\hat{y}_{ij}$  is the predicted probability for the same instance.

The proximal term, which penalizes the discrepancy between the local model's weights ( $\omega_i$ ) and the global model's weights ( $\omega_{global}$ ):

$$\mathcal{L}_{FedProx}(\omega_i) = \mathcal{L}_{local}(D_i, \omega_i) + \frac{\mu}{2} \|\omega_i - \omega_{global}\|^2$$

where  $\mu$  is a hyperparameter controlling the influence of this term, the proximal term ensures that local models do not diverge significantly from the global model, which can be crucial when clients have non-iid data.

#### F. INITIALIZE GLOBAL MODEL

A global CNN model is initialized with a set of initial weights, typically set randomly or using pre-trained weights from a similar task. This model serves as the baseline for federated training and will be iteratively updated based on client contributions.

#### G. FEDERATED TRAINING FOR $R$ ROUNDS

The training process is executed over  $R$  rounds, following these sub-steps:

1. **Distribute Global Weights to Clients:** The global model weights are sent to each client, allowing them to start training from the same baseline.
2. **Local Training at Client  $i$ :** Each client  $i$  trains its local model on its dataset using the FedProx loss function defined earlier. This involves updating the model weights based on local data and minimizing the FedProx loss over several epochs:

$$\omega_i^r = \omega_i^{r-1} - \eta \nabla \mathcal{L}_{FedProx}(\omega_i)$$

where  $\eta$  is the learning rate, and  $\nabla \mathcal{L}_{FedProx}(\omega_i)$  is the gradient of the FedProx loss.

3. **Send Local Model Updates:** After local training, clients send their updated weights back to the server. This communication can be limited to the changes in weights to save bandwidth.
4. **Federated Averaging (FedAvg):** The server aggregates the weights received from all clients to update the global model. This is done by averaging the weights:

$$\omega_{global}^r = \frac{1}{K} \sum_{i=1}^K \omega_i^r$$

where  $K$  is the number of clients, this method assumes that all clients contribute equally. However, weighted averages can also be used based on the amount of data each client holds

To calculate FedAvg we followed below steps:

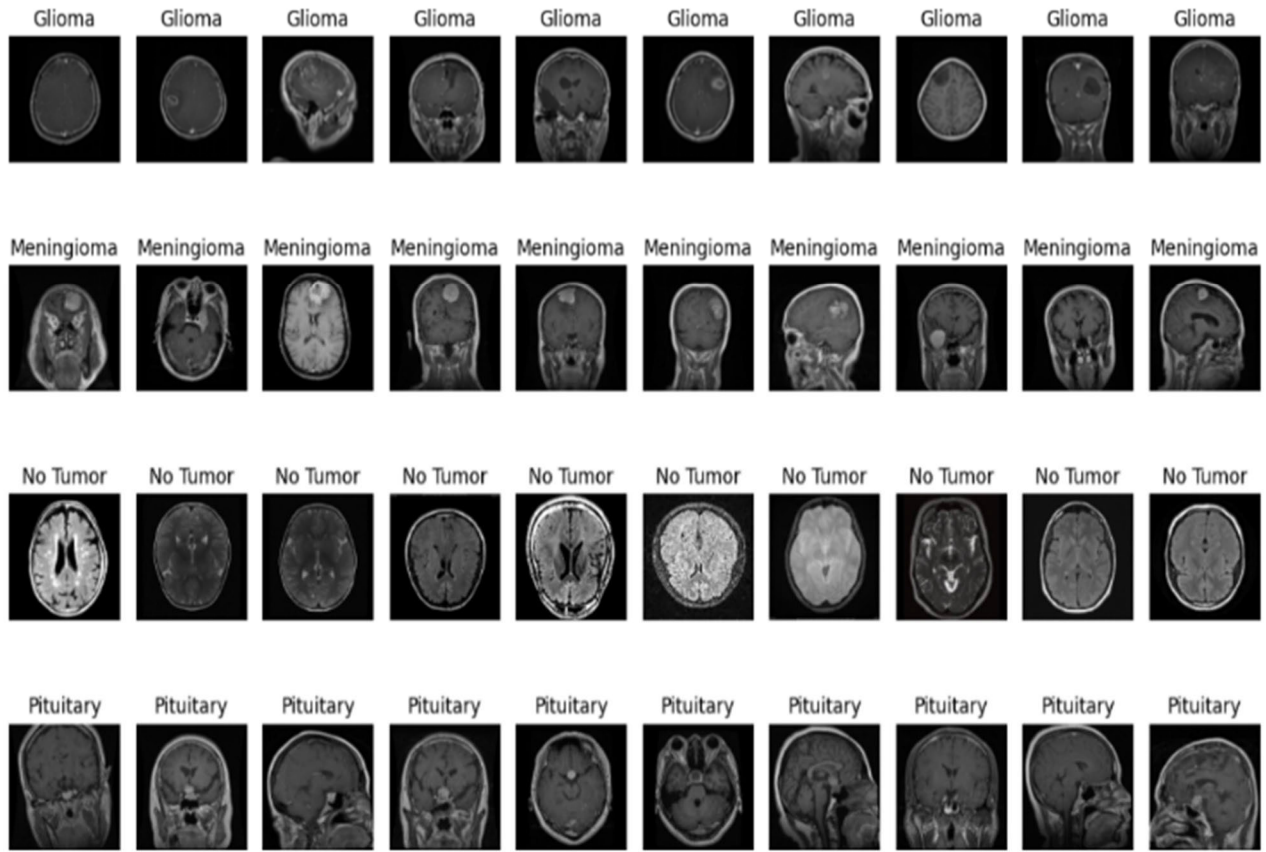


FIGURE 3. Four categories: glioma, meningioma, no tumor, and pituitary.

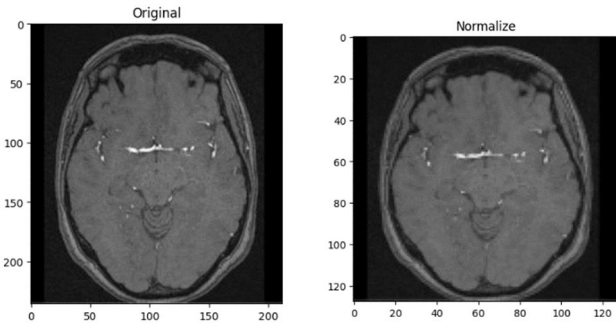


FIGURE 4. Original and preprocessed image.

- Initially, we start with randomly initialized global model weights. We assume the initial global weights at Round 0:

$$\omega_{global}^0 = [0.5, 0.5, 0.5]$$

- These weights are sent to all clients. Each client **trains the model locally** for 10 epochs. The **updated local model weights** after training depend on:
  - FedProx Loss:** Prevents extreme weight divergence.
  - Local Data Distribution:** Clients update weights based on their dataset.

TABLE 2. Updated local model weights in round 1.

| Client   | Local Model Weights $w_i^1$ after Round 1 |
|----------|-------------------------------------------|
| Client 1 | [0.55, 0.52, 0.53]                        |
| Client 2 | [0.57, 0.51, 0.54]                        |
| Client 3 | [0.58, 0.49, 0.52]                        |
| Client 4 | [0.56, 0.50, 0.55]                        |
| Client 5 | [0.54, 0.53, 0.51]                        |

As shown in table 2, we have used 5 clients ( $K = 5$ ), each updating their weights:

- To Calculate First Global Weight, we used FedAvg Aggregation. After training, the server aggregates weights using FedAvg:

$$\omega_{global}^1 = \frac{1}{K} \sum_{i=1}^K \omega_i^1$$

1) Compute Averaged Weights for Round 1  
First weight component:

$$\frac{1}{5} (0.55 + 0.57 + 0.58 + 0.56 + 0.54) = 0.56$$

**TABLE 3.** Global model weights per rounds.

| Round | Global Model Weights $\omega_{global}^t$ |
|-------|------------------------------------------|
| 1     | [0.56,0.51,0.53]                         |
| 2     | [0.60,0.54,0.55]                         |
| 3     | [0.64,0.57,0.58]                         |
| 4     | [0.67,0.60,0.61]                         |
| 5     | [0.71,0.63,0.65]                         |
| 6     | [0.75,0.67,0.69]                         |
| 7     | [0.79,0.70,0.72]                         |
| 8     | [0.82, 0.73, 0.75]                       |
| 9     | [0.86,0.76,0.78]                         |
| 10    | [0.89,0.79,0.81]                         |

**Second weight component:**

$$\frac{1}{5}(0.52 + 0.51 + 0.49 + 0.50 + 0.53) = 0.51$$

**Third weight component:**

$$\frac{1}{5}(0.53 + 0.54 + 0.52 + 0.55 + 0.51) = 0.53$$

Thus, the **new global model weights after Round 1:**

$$\omega_{global}^1 = [0.56, 0.51, 0.53]$$

4. Now, these updated global weights  $\omega_{global}^1$  are sent back to the clients for the next round and step 3 will be repeated for Round 2 to Round 10. Finally, the updated global weights are obtained below.

Once the final global weights are computed at **Round 10** as shown in table 3, they are used to predict on the test dataset.

#### H. EVALUATE GLOBAL MODEL

$$\begin{aligned}
 \text{Accuracy} &= \frac{\text{Number of correct prediction}}{\text{Total number of prediction}} \\
 &= \frac{TP + TN}{TP + TN + FP + FN} \\
 \text{Precision} &= \frac{TP}{TP + FP} \\
 \text{Recall} &= \frac{TP}{TP + FN} \\
 F1 &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}
 \end{aligned}$$

#### I. PLOT METRICS ACROSS ROUNDS

To visualize the global model performance over training rounds, metrics are plotted. This helps in identifying trends and the effectiveness of the FL approach in training the model

#### J. DEPLOY FINAL MODEL

After completing the federated training over R rounds, the final global model is ready for deployment.

The **Federated Averaging (FedAvg)** algorithm is a commonly used technique in FL, particularly useful for collaboratively training a global model across multiple parties while securing data privacy. In the context of **BTC**, FedAvg ensures that a central server coordinates training without requiring clients to share their sensitive MRI data. The process works as follows:

1. A local model is trained on a separate subset of data by each client.
2. Once a communication round is complete, the clients send their locally trained model weights to the server.
3. The server then aggregates the updates by averaging their weights to get a new global model, which is then distributed back to the clients for the next round of training.
4. Mathematically, if  $\omega_i^r$  represents the model weights for client  $i$  after round  $r$ , the updated global weights  $\omega_{global}^r$  are computed as:

$$\omega_{global}^r = \frac{1}{K} \sum_{i=1}^K \omega_i^r$$

where  $K$  is the total number of clients, the key idea behind FedAvg is that the global model learns from the diverse data distributions across clients, resulting in a model that generalizes well without the need for centralized data storage. This is particularly beneficial for tasks like brain tumor classification, where medical data privacy is critical. However, FedAvg assumes that all clients contribute equally and does not explicitly handle situations where clients have heterogeneous (non-IID) data distributions, leading to potential issues when data across clients varies significantly.

The **Federated Proximal (FedProx)** algorithm builds upon FedAvg to address the challenges of client heterogeneity, especially when the data distribution is non-IID across clients. In FL for brain tumor classification, this can occur when MRI scans differ significantly between hospitals or regions. FedProx introduces a **proximal term** in the loss function to penalize large deviations of local model weights from the global model. This ensures that even when clients have vastly different data distributions, their local models remain aligned with the global model during training. The FedProx loss function at client  $i$  is given by:

$$\mathcal{L}_{\text{FedProx}}(\omega_i) = \mathcal{L}_{\text{local}}(D_i, \omega_i) + \frac{\mu}{2} \|\omega_i - \omega_{global}\|^2$$

Here,  $\mathcal{L}_{\text{local}}(D_i, \omega_i)$  is the local loss function (categorical cross-entropy in the case of classification),  $\omega_i$  are the local weights,  $\omega_{global}$  are the global model weights, and  $\mu$  is a positive coefficient controlling the weight of the proximal term. The proximal term  $\frac{\mu}{2} \|\omega_i - \omega_{global}\|^2$  ensures that the local model updates do not diverge too much from the global model, thereby stabilizing training in non-IID environments. This is particularly important in brain tumor classification tasks where MRI scans may be significantly different between clients, yet the global model must still

**ALGORITHM :**

- 2) Step 1: Dataset Preparation and Initialization
  - Load and Pre-process Data:*
    - Load the brain tumor MRI dataset
    - Resize images to  $128 \times 128$  pixels and normalize them
    - One-hot encode class labels (e.g., glioma = [1, 0, 0, 0]).
  - Split Data Among Clients*
    - Split the dataset into  $K$  parts (one for each client).
    - Ensure each client gets approximately the same amount of data.
- 3) Step 2: Model Definition and Loss Function
  - Define CNN Model*
    - Build a CNN with convolutional layers, pooling layers, activation (ReLU), and a softmax output layer for classification into four categories
  - Define FedProx Loss*
    - Use categorical cross-entropy for local loss.
    - Add a proximal term to penalize the difference between local weights and global weights to handle client heterogeneity:
 
$$\mathcal{L}_{\text{FedProx}}(\omega_i) = \mathcal{L}_{\text{local}}(D_i, \omega_i) + \frac{\mu}{2} \|\omega_i - \omega_{\text{global}}\|^2$$
 $\mu$  controls the weight of this term.
- 4) Step 3: Federated Training Loop
  - Initialize Global Model*
    - Create a global CNN model with initial weights.
  - Federated Training for  $R$  Rounds*
    - Distribute Global Weights to Clients*
      - Send global model weights to each client.
    - Local Training at Client  $i$* 
      - Each client trains its local model on its data using the FedProx loss function.
    - Send Local Model Updates*
      - After training, clients send their updated weights back to the server.
    - Federated Averaging (FedAvg):*
      - The server averages the weights from all clients to update the global model:
 
$$\omega_{\text{global}}^r = \frac{1}{K} \sum_{i=1}^K \omega_i^r$$
- 5) Step 4: Global Model Evaluation
  - Evaluate Global Model*
    - Test the global model on a testing dataset after each round.
- 6) Step 5: Visualization and Final Deployment
  - Plot Metrics Across Rounds*
  - Deploy Final Model*

be generalized effectively. By constraining the local models, FedProx helps the global model converge more robustly, mitigating the risk of poor performance due to client data heterogeneity.

## V. RESULT AND DISCUSSION

### A. GLOBAL MODEL ACCURACY

The model performance for Round 1 is 0.8899, which is a reasonable start. It tells that the model for the assignment is quite good as it is. It was observed that the accuracy rate remains relatively stable across Rounds 2 to Round 4, but it continues to rise gradually to 0.9176 in Round 4.

The accuracy at Round 5 jumps to 0.9413 to indicate that the model has been trained. Starting from Round 6 to Round 10, the accuracy rises again as much as in Round 10, with an accuracy of 0.9719. In general, there is a systematic improvement in the accuracy over the rounds, especially in the first five rounds. The following rounds do not present a significant growth, which is quite typical since the model is close to its final conclusions as shown in figure 5.

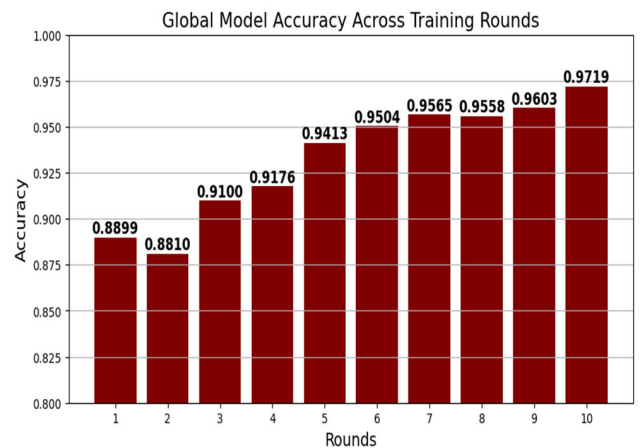


FIGURE 5. Global model accuracy.

### B. GLOBAL PRECISION

As shown in figure 6, the precision of this system is 0.8186 in Round 1, which is an indication of a fairly large number of false positives. In Round 2, precision is 0.8788 and it is

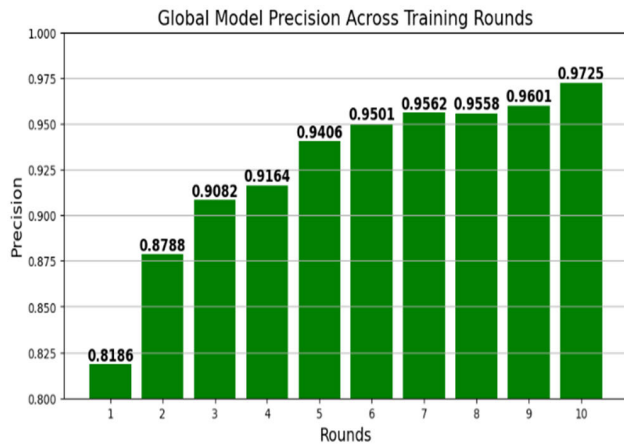


FIGURE 6. Global model precision.

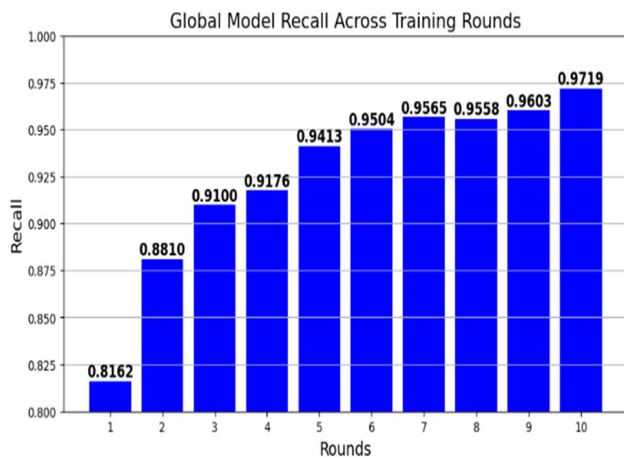


FIGURE 7. Global model recall.

quite visible that the results have become better. The precision increases gradually round by round, standing at 0.9164 in Round 4 and 0.9406 in Round 5. The precision of the system is at its highest at 0.9725 in Round 10, meaning that the system has been able to minimize the false positives. This is evidenced by the rising precision as the model trains as it becomes spot on in predicting positive cases while making fewer wrong decisions.

### C. GLOBAL RECALL

The recall begins at 0.8162 in the first round and is almost equivalent to the precision value, which shows the model's ability to classify positives without generating too many false alarms as shown in figure 7. From Round 2 to Round 4, the recall increases gradually and gets to 0.9176 in Round 4. Recall capacity also remains on the rise and rises dramatically to 0.9504 in Round 6 and then reaches its highest level at 0.9719 by Round 10. This is evidence that the model is improving at recognizing the true positives and not compromising the ability to avoid false negatives.

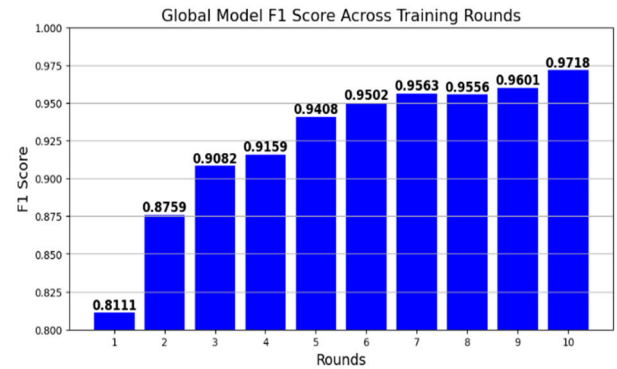


FIGURE 8. Global mode F1 score.

### D. GLOBAL F1 SCORE

In figure 8, the F1 score comes to 0.8111 in Round 1, and it is slightly lower than the precision and recall but quite close to the mode's initial performance metrics. The F1 score for Round 2 is 0.8759, which is an increase from the previous round. Thus, by Round 5, the F1 score has risen to 0.9408 and is evidence of good performance both in terms of true positive prediction and minimal false positives. The F1 score keeps on rising, reaching 0.9718 in Round 10, this shows that the model is well equipped to balance between precision and recall. The F1 score gives a similar pattern to the other metrics, with a steep increase at round 5 and then a slow and steady rise till the end. This implies that at the later rounds, the model is not only increasing its efficiency in identifying the positive cases but also reducing the wrong identification.:

Table 4 presents the progression of a global model's performance over 10 rounds of training or evaluation, with the key metrics values. Let's break down and analyze each round and the trends observed for these metrics.

### E. ANALYSIS AND OBSERVATION

1. *Steady Improvement Across All Metrics:* The trend for all the key metrics is upward, with even more dramatic increases in the early rounds. There is a significant change in the model by Round 5 from Round 1, with a further marginal increase up to Round 10. So, this tells that the model is learning good.

2. *Precision-Recall Balance:* Both the precision and recall values are quite similar for all the rounds, which means that the model does not predict many false positive or false negative cases. This is important, especially in tasks such as medical image classification, where both false negatives and false positives are expensive.

3. *Significant Jump in Mid-Rounds:* We can see a marked improvement in the performance between Round 4 and Round 5 in terms of accuracy and F1 score. This could suggest that the model has arrived at some critical point at which the generalization capability was enhanced significantly. From Round 5 onwards, the improvements are not much, this is an indication that the model is converging gradually.

**TABLE 4.** Global model evaluation metrics across rounds.

| Rounds | Global Model Accuracy | Global Precision | Global Recall | Global F1 Score |
|--------|-----------------------|------------------|---------------|-----------------|
| 1      | 0.8899                | 0.8186           | 0.8162        | 0.8111          |
| 2      | 0.881                 | 0.8788           | 0.881         | 0.8759          |
| 3      | 0.91                  | 0.9082           | 0.91          | 0.9082          |
| 4      | 0.9176                | 0.9164           | 0.9176        | 0.9159          |
| 5      | 0.9413                | 0.9406           | 0.9413        | 0.9408          |
| 6      | 0.9504                | 0.9501           | 0.9504        | 0.9502          |
| 7      | 0.9565                | 0.9562           | 0.9565        | 0.9563          |
| 8      | 0.9558                | 0.9558           | 0.9558        | 0.9556          |
| 9      | 0.9603                | 0.9601           | 0.9603        | 0.9601          |
| 10     | 0.9719                | 0.9725           | 0.9719        | 0.9718          |

4. Convergence and Saturation: At Round 8, the trends of the accuracy and F1 score become relatively stable, and the increases are insignificant between Rounds 9 and 10. This is expected as the model maximizes in a way that additional training may not necessarily lead to the highest result. This means it has become quite stable at the end of Round 10 with accuracy and F1 performance of more than 0.97.

## VI. COMPARATIVE ANALYSIS

The table 5 shows various models for the classification of brain tumors with their accuracies. The proposed analysis starts with BWT and ROBPCA, the method used by Chappidi Sree Teja Reddy et al. in 2024. The BWT and ROBPCA obtained accuracies of 81.50% and 84%, respectively. These reduced models are easy to implement but may fail in more challenging imaging situations, which shows the need for more elaborate methods in tumor categorization.

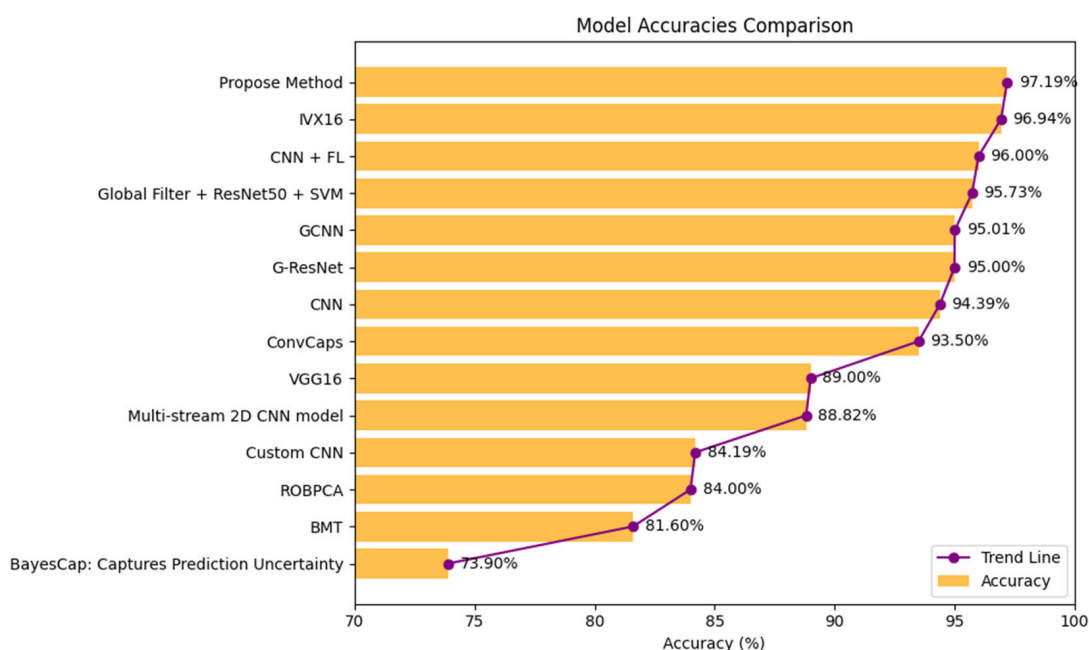
Ullah et al. [26] used Gabor filters along with ResNet50 and SVM, where the accuracy of the system was found to be 95.73%, respectively. On the other hand, this hybrid method takes advantage of the feature extraction property of DL models, but at the same time, it adds computational complexity. Ravinder et al. [27] then improved this with a Graph-based Convolutional Neural Network (GCNN) which yielded an accuracy of 95.01%. The GCNN model is an extension of GNN and CNN, for non-Euclidean data, but it is expensive in terms of training. Hossain et al. [29] have come up with a new model with an accuracy of 96.94%. This approach showed that using deep learning methods for classification is viable and efficient even though the size of the dataset remains a problem that affects most deep learning applications. Das et al. [30] used a basic CNN model, which gave an accuracy of 94.39%.

Liu D et al. proposed the G-ResNet model which achieved 95% accuracy and demonstrated capabilities of deep learning based classification using residual networks. Ge C et al. also created a Multi-stream 2D-CNN model with 88.82% accuracy that utilized feature extraction with multiple input streams. Afshar P et al. had lower accuracy with 73.90% while implementing BayesCap, a model aimed at capturing prediction uncertainty, highlighting issues of uncertainty quantification for classification. Cheng Y et al. on the other hand used the ConvCaps model and reported an accuracy of 93.50%, illustrating the effectiveness of capsule networks in retaining spatial hierarchies. Further, Abiwiananda N et al. built a Custom CNN model that had 84.19% accuracy, showcasing flexibility of architectures based on CNNs. Naser M A et al. performed well with VGG16, a deep convolutional network, and reported an accuracy of 89%. S. Bhadauriya et al. reported using CNN integrated with Federated Learning (FL) and reported a high accuracy of 96% showing the effectiveness of these new learning methods.

As shown in the figure 9, the FedAvg+FedProx model proposed in this work achieved an accuracy of 97.19%, which is the highest accuracy compared with other models presented in this work or in the literature. FL Unlike the other architectures, FedAvg+FedProx uses a decentralized learning approach to address the challenge of client cooperation through model training without transferring data. For overall model training, FedAvg+FedProx approaches data heterogeneity in clients, which can be hospitals or medical centers, by using the FedProx algorithm, which makes the training process stable even for clients with very distinct data distribution. This enhancement is of particular value as the model can be applied across a wide range of sources of data and is, therefore, stronger than other models based on a single source of data.

**TABLE 5.** Accuracy comparison.

| Ref No                | Authors                         | Model                                     | Accuracy     |
|-----------------------|---------------------------------|-------------------------------------------|--------------|
| [26]                  | Ullah et al.                    | Gabor Filter + ResNet50 + SVM             | 95.73%       |
| [27]                  | M. Ravinder et al.              | GCNN                                      | 95.01%       |
| [28]                  | Chappidi Sree Teja Reddy et al. | BWT                                       | 81.50%       |
| [28]                  | Chappidi Sree Teja Reddy et al. | ROBPCA                                    | 84%          |
| [29]                  | S. Hossain et al.               | IVX16                                     | 96.94%       |
| [30]                  | S. Das et al.                   | CNN                                       | 94.39%       |
| [31]                  | Liu D et al.                    | G-ResNet                                  | 95%          |
| [32]                  | Ge C et al.                     | Multi-stream 2D-CNN model                 | 88.82%       |
| [33]                  | Afshar P et al.                 | BayesCap: captures prediction uncertainty | 73.90%       |
| [34]                  | Cheng Y et al.                  | ConvCaps                                  | 93.50%       |
| [35]                  | Abiwinanda N et al.             | Custom CNN                                | 84.19%       |
| [36]                  | Naser M.A et al.                | VGG16                                     | 89%          |
| [37]                  | S. Bhadauriya et al.            | CNN + FL                                  | 96%          |
| <b>Proposed Model</b> |                                 | <b>CNN+FedAvg+FedProx</b>                 | <b>97.19</b> |

**FIGURE 9.** Accuracy comparison of proposed model.

## VII. LIMITATIONS AND FUTURE WORK

This study carries with it some limitations, such as using a highly specific and constrained database offered by Kaggle which does not represent real life variations in MRI scans, patient demographic, and imaging procedures, thus affect-

ing generalizability. Furthermore, the scope is still an issue since the FL framework was implemented with a small group of clients compared to what would be found in practice at various healthcare institutions that differ in terms of network penetration and hardware availability. Implementing

FL, particularly with CNN based models, also incurs significant computation and communication expenses which, as is known, are rather difficult to meet in resource constrained environments. And finally, the model is clinically has not been applied against actual hospital data, which is one of the crucial steps towards making it reliable in real life medical practices.

The future work would be attempting to lessen the communication and computation overload that is present utilizing model compression methods along with optimized communication techniques that would increase the feasibility of implementing FL-based CNN models in a constrained environment. Moreover, clinical validation and real-life testing are necessary steps that must be done to assess how the model would perform with real hospital data, making it more generalizable to varying patient populations and imaging conditions, so as to enhance its reliability. An integrated approach using other modalities alongside MRI scans such as clinical notes or case history of the patient can increase the diagnostic accuracy through the use of complementary data and improve it. Also, the use of explainable AI (XAI) will increase the level of understanding of the medical imaging decision-making model so that it becomes easy for the medical professionals to accept and trust its outcomes, which in turn makes it possible to meet the legal and ethical responsibilities of using AI in medical imaging.

## VIII. CONCLUSION

This BTC work implemented using FL shows major improvement in the model performance and its ability to generalize with the help of the Proposed Fed+FedProx approach. The work then overcomes the challenges of data privacy, heterogeneity, and distributed learning by extending and combining FedAvg and FedProx. The Fed+FedProx model proposed in this paper is better than conventional architectures. The employment of FedAvg helps in the formation of updates from the participating clients to the server reducing learning bias and improving the learning process while at the same time maintaining the privacy of the patient's data. However, FedProx adds a proximal term to leverage the model's capacity to learn from data that is not Independent and Identically Distributed (non-IID) across different clients.

The Proposed Fed+FedProx model achieved 97.19% accuracy and 97.18% F1 score, which confirms the success of FL in the medical imaging domain. This approach not only enhances the classification performance but also facilitates better clinical decisions through accurate and private prediction. Altogether, the results of this work indicate that FL can be applied successfully for improvements of the brain tumor classification systems using FedAvg and FedProx algorithms. This study presents a foundation for future works and real implementations in medical imaging, thus opening new possibilities for improving the existing and developing new mechanisms of Machine Learning in the healthcare sector.

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